

Introduction to Text-as-Data & Classifying Texts with LLMs

Dr. Fabienne Lind (University of Vienna)



Wednesday | 22.07.2025

Summer School for Women* in
Political Methodology (2026)

Text

Workshop objectives

- Getting to know (basic) procedures of computational text analysis
 - **Apply** basic text as data methods
 - Insights into **practical challenges**
 - **Critical reflection** on the method
 - **Inspiration** for your own projects
-

Course philosophie

Topics are covered with

- Lecture style input
- Group discussions
- Guided coding sessions

Interrupt me, ask all kinds of questions

Course materials

<https://github.com/fabiennelind/text-as-data-in-R>

Agenda

09:00 - 09:15	Orga
09:15 - 12:30	Introduction to Text-as-Data
12:30 - 13:30	Lunch Break
13:30 - 16:45	Classifying Texts with LLMs
16:45 - 17:15	Feedback on your projects (Optional)

Feedback on your projects (optional)

When:

- Today after class 16:45 to 17:15

What:

- Informal opportunities to talk about your text analysis use case and (initial) design (plan), Research question, Data, Collection, Current struggles
 - Receive feedback
-

About me

Fabienne Lind

- Communication Scientist
- Methods development: Multilingual computational text analysis, comparative analysis
- Substantial focus: Political communication, environmental communication
- fabienne.lind@univie.ac.at



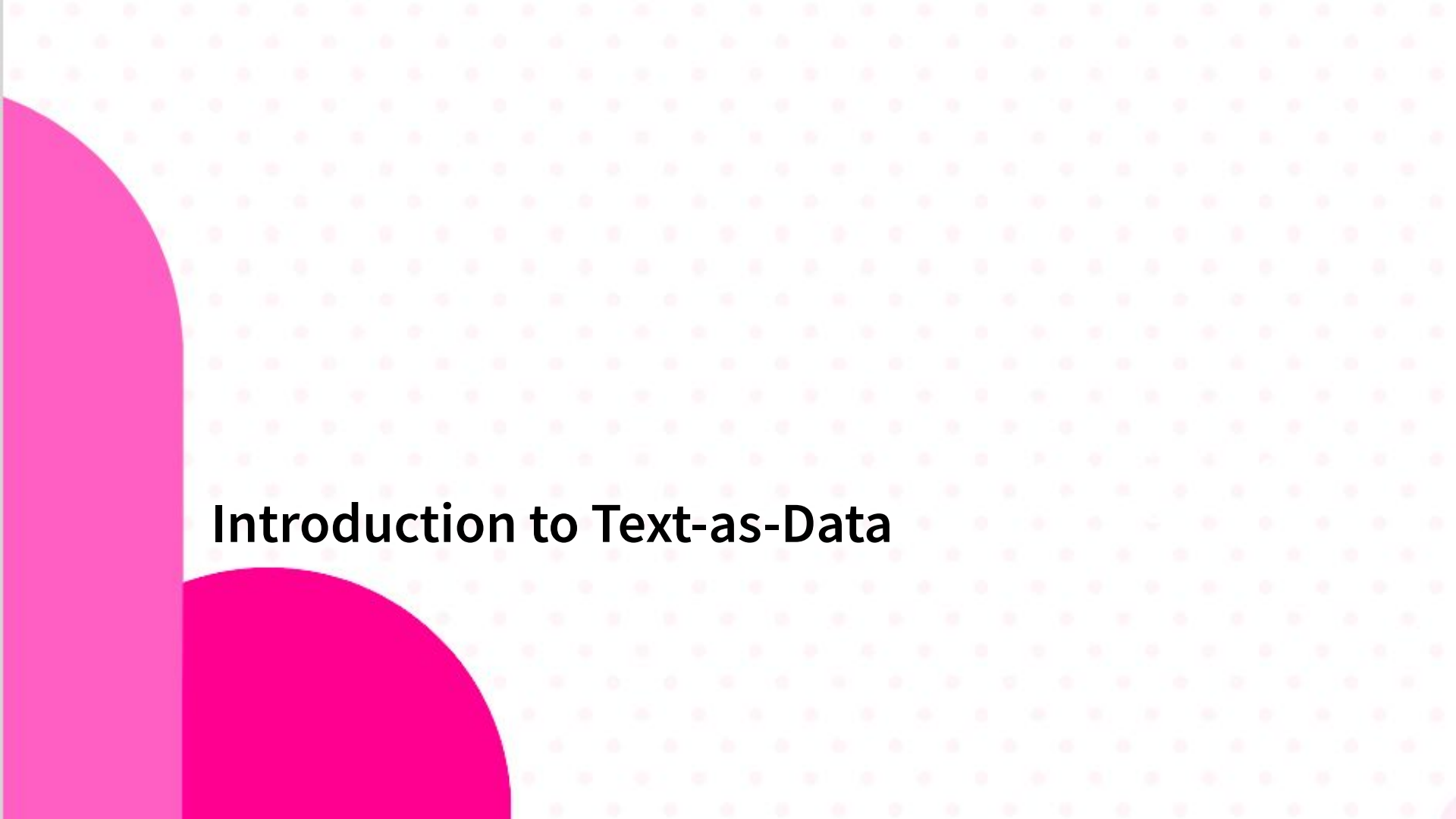
Your turn :)



Go to <https://www.menti.com>

Enter the code: 8502 3495

<https://www.menti.com/al7epayw63gd>

The background features a light pink grid of small dots. On the left side, there are two large, overlapping magenta shapes: a vertical rectangle and a semi-circle.

Introduction to Text-as-Data

Agenda

09:15 - 10:45	Dictionary methods Overview + R Session
10:45 - 11:15	Coffee break
11:15 - 12:30	Supervised machine learning Overview + R Session

Motivations to analyse text automatically

Limitations of Manual Coding:

- Slow
- Limited scale
- Reliability issues



Motivations to analyse text

- Huge volumes of digital available information
- These data can be useful to research media contents, communicators, effects, and user interactions
- Describe, explain, 'predict' feelings, attitudes and behaviour of individuals, groups, societies

Example 1: Contents and Effects

Research Question: How prevalent is each frame in NGOs' tweets?

- Approach:** Dictionary

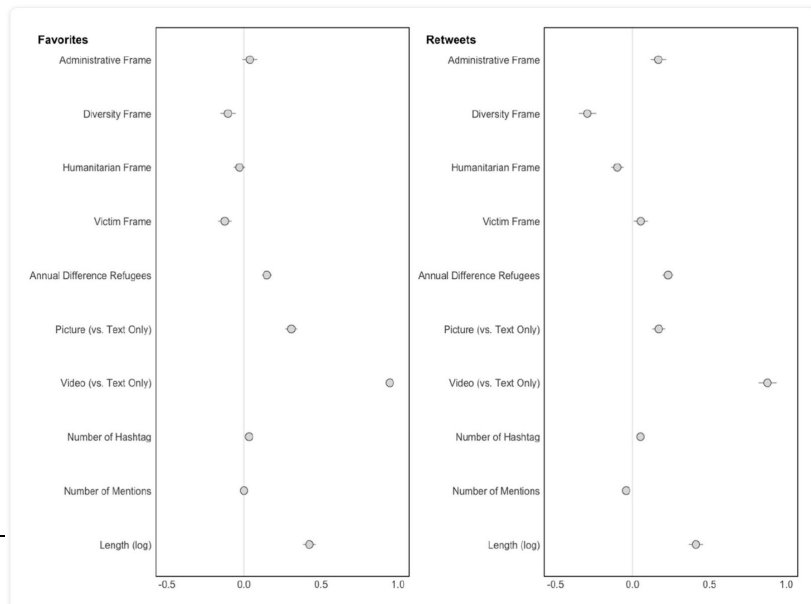
Table 3. Frame frequencies in Twitter communication of top 10 humanitarian NGOs.

Frame	Frequency (percent)
Administrative frame	2301 (11.78)
Diversity frame	2028 (10.39)
Humanitarian frame	4411 (22.59)
Victim frame	3075 (15.75)

Note. $N=19,528$. Part of the sample did not contain any of the four frames measured.

Dimitrova, D., Heidenreich, T., & Georgiev, T. A. (2022). The relationship between humanitarian NGO communication and user engagement on Twitter. *New Media & Society*, 14614448221088970.

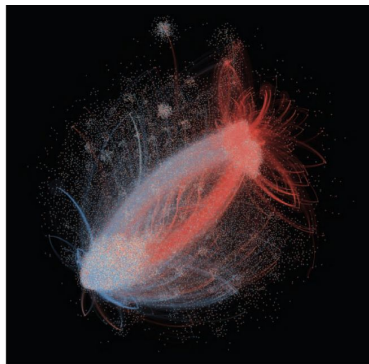
- Research Question:** What reactions do the frames provoke?



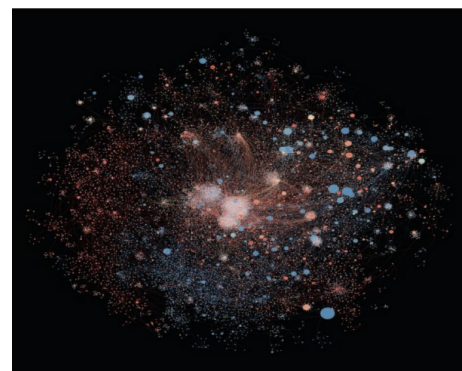
Example 2: Data selection for network analysis

Data selection requirement: Which of the 12 defined topics was mentioned in which tweet?

- **Approach:** Dictionary



2012 Presidential Election



2013 Super Bowl

Barbera et al., 2015, Fig. 3

Purpose of obtaining measures for a large number of documents

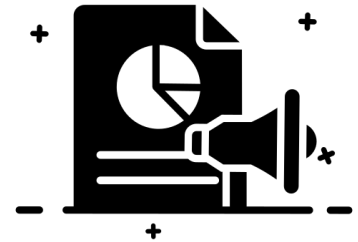
With rich data, more advanced filtering becomes possible

- Examples: linkage studies combination with media usage data; speech data for various actors



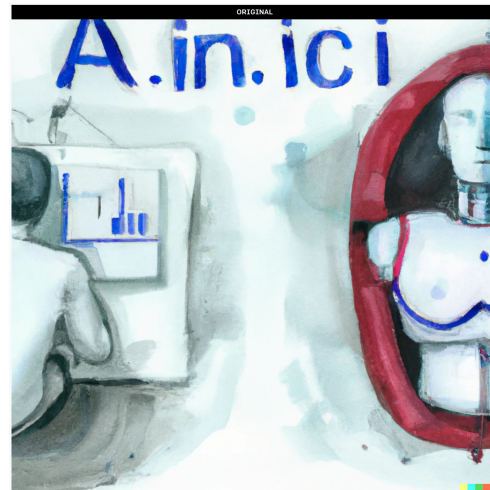
Evidence-based policy making

- Making the opinion of populations visible,
- Holding those in power accountable



The end of manual coding?

- Augmenting not replacing (Grimmer & Stewart, 2013)
- Human input for quality control:
 - select, monitor, and test on the level of data, inputs, process, outputs



Some major challenges when working with large corpora

Big data, big bias?

The end of theory?

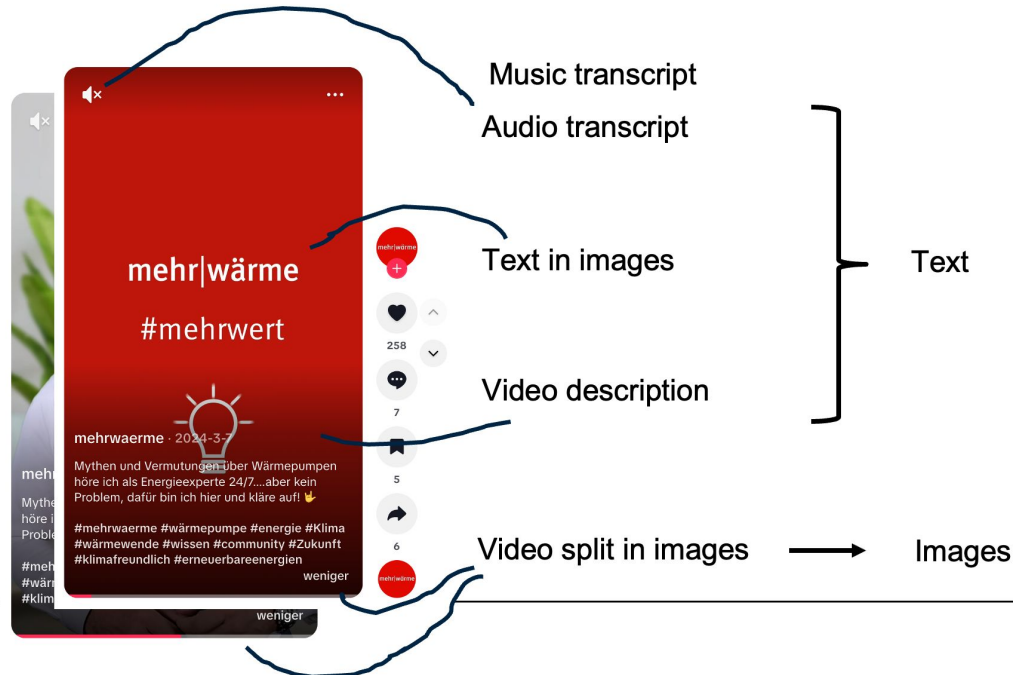
Generalizing from online to offline behavior

Ethical concerns

Text types (Examples)

- Social media posts and comments
 - Transcribed speeches
 - Wikipedia articles
 - Open answer questions in surveys
 - CVs
 - Transcripts of interviews
 - AI generated texts
 - Etc.
-

Text types (What about videos?)



Your projects

- What are your applications of text analysis?
- What text types do you study?
- Where do you see challenges?



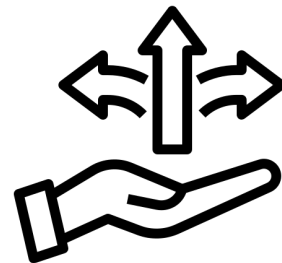
Dictionaries

Dictionary method

Definition: List of terms (words or phrases) that indicate a category or concept

Application: Count occurrences of dictionary terms within documents to score them on the concept

Approach: Theory-driven, use when a category is known (= defined by researcher)



Example: Dictionary for the concept “EU countries”

```
eu_countries <- c("Austria", "Belgium", "Bulgaria", "Croatia", "Cyprus",  
"Czech Republic", "Denmark", "Estonia", "Finland", "France",  
"Germany", "Greece", "Hungary", "Ireland", "Italy", "Latvia",  
"Lithuania", "Luxembourg", "Malta", "Netherlands", "Poland",  
"Portugal", "Romania", "Slovakia", "Slovenia", "Spain", "Sweden")
```



Is the list exhaustive to match all mentions of EU countries in text?

Example: Dictionary for the concept “EU countries”

```
eu_countries <- c("Austria", "Belgium", "Bulgaria", "Croatia", "Cyprus",  
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"Germany", "Greece", "Hungary", "Ireland", "Italy", "Latvia",  
"Lithuania", "Luxembourg", "Malta", "Netherlands", "Poland",  
"Portugal", "Romania", "Slovakia", "Slovenia", "Spain", "Sweden")
```



Is the list exhaustive to match all mentions of EU countries in text?

- Abbreviations (AT), different common spellings (e.g., Czechia), historical data (UK)

Dictionary use case 1: Filter Data

Example szenario:

- You have a text corpus about inflation.
 - You are interested in analysing articles that discuss the inflation topic in the context of the next national election.
 - You construct a dictionary to filter the data.
-

Dictionary use case 1: Filter Data

- All texts are about **inflation**. What words indicate that some are also related to a **national election**?

The Swiss National Bank became the first central bank among advanced economies to declare victory over the post-pandemic surge in **inflation**, cutting its key interest rate by 0.25 percentage point to 1.5 percent and hinting at more cuts in the coming months.

Politico.eu, March 21, .2024

“Even as incomes are going up and the economy is doing well and **inflation** is coming down, people can’t buy homes,” said Daryl Fairweather, chief economist at the brokerage Redfin. “That’s like the biggest problem for Biden because it’s not one that he can solve.”

AP News, March 15.2024

Global Fuel Prices Are Surging With Supply Risks Ahead

Bloomberg, March 12.2024

- US pump prices 60% higher than 2020, when Biden was elected
- Global oil demand and refinery runs are both forecast to rise

Dictionary use case 1: Filter Data

- Keep only articles where your words appear

“Even as incomes are going up and the economy is doing well and inflation is coming down, people can’t buy homes,” said Daryl Fairweather, chief economist at the brokerage Redfin. “That’s like the biggest problem for Biden because it’s not one that he can solve.”

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Dictionary use case 2: Measure classes of category

Example szenario:

- You have a corpus about inflation and the next national election
- You design dictionaries to know which article is about **housing** and which is about **energy**



Dictionary use case 2: Measure classes of category

- What words indicate that a text is about the **housing** topic?
- What words indicate that a text is about the **energy** topic?

“Even as incomes are going up and the economy is doing well and inflation is coming down, people can’t buy homes,” said Daryl Fairweather, chief economist at the brokerage Redfin. “That’s like the biggest problem for Biden because it’s not one that he can solve.”

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Dictionary use case 2: Measure classes of category

- What words indicate that a text is about the **housing** topic?
- What words indicate that a text is about the **energy** topic?

“Even as incomes are going up and the economy is doing well and inflation is coming down, people can’t buy **homes**,” said Daryl Fairweather, chief economist at the brokerage Redfin. “That’s like the biggest problem for Biden because it’s not one that he can solve.”

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Application: Classify actor salience

Objective: salience of women and men migrant's in the news across time and outlet

- Dictionary approach to measure mentions of women and men migrants in German news articles



Example: Classify actor salience

Keyword list more complex than initially thought

```

349
350
351
352 person_f_migrant_endIn_regex = c("[Aa]syntantin", "[Aa]syntawerberin", "[Aa]
353 "[Zz]uwanderin", "[Ee]inwanderin", "[Gg]a
354 "[Aa]usländische\\w{0,2}\\s[Bb]ürgerin",
355
356
357 f_relation_regex = c("[Mm]eine\\w{0,2}\\s[Ff]rau(en)\\C\\s\\W)", "[Mm]ein\\
358 "[Dd]eine\\w{0,2}\\s[Ff]rau(en)\\C\\s\\W)", "[Dd]ein\\w{
359 "[Ss]eine\\w{0,2}\\s[Ff]rau(en)\\C\\s\\W)", "[Ss]ein\\w{
360 "[Ii]hr\\w{0,2}\\s[Ff]rau(en)\\C\\s\\W)", "[Ii]hr\\w{0,2
361 "[Uu]ns\\w{0,2}\\s[Ff]rau(en)\\C\\s\\W)", "[Uu]ns(er)re
362 "[Ee]u(er)re\\w{0,2}\\s[Ff]rau(en)\\C\\s\\W)", "[Ee]u(er
363
364
365
366 f_relation_regex <- paste(apos_closed, f_relation_regex, "", sep = "") #add
367 df_f_relation_regex <- data.frame(f_relation_regex) #as. dataframe

```

Table 2. Dictionary subcategories for the measurement of the concepts: migrant women's and migrant men's salience.

Subcategory name	Description	Measured concepts (examples)	
		Migrant women's salience	Migrant men's salience
1. Impersonal designation	General person nominations for migrants	(immigrant, asylum seeker, etc.) with female or gender-neutral word endings (-In, -in, etc.)	(immigrant, asylum seeker, etc.) with generic masculine form or gender-neutral word endings (-In, -in, etc.)
2. Origin: Single words	Nominations that refer to the territorial origin of a person ^a	(French, African, Syrian, etc.) with female or gender-neutral word endings (-In, -in, etc.)	(French, African, Syrian, etc.) with generic masculine form or gender-neutral word endings (-In, -in, etc.)
3. Origin: Combinations of different word groups ^b			
a	General gender-related person nominations + from + general territorial denominations ^c	(women, girl, mother, sister) + from + (France, Africa, Syria, etc.)	(man, boy, father, brother) + from + (France, Africa, Syria, etc.)
b	General territorial denominations (adjectives) ^c + General gender-related person nominations	(French, African, Syrian, etc.) + (women, girl, mother, sister)	(French, African, Syrian) + (man, boy, father, brother)
4. In relation	Relational expressions: possessive pronouns + General gender-related person nominations	(my, your, her, his, our, yours, their) + (women, girl, mother, sister)	(my, your, her, his, our, yours, their) + (man, boy, father, brother)
5. As phrase	Phrases	e.g., "women and children"	e.g., "men and children"
6. Other expressions		e.g., "women from abroad"	e.g., "men from abroad"

As general notes, the dictionary includes the singular and plural version of all words, word endings (e.g., for prepositions) consider the different cases used in the German language.

^aDownloaded from the CLDR (Unicode Common Locale Data Repository) <http://cldr.unicode.org/>, which holds standard name translations of countries and regions (version v33.1).

^bMeasured at the sentence level.

^cManually compiled by a native speaker for all CLDR territorial denominations, which the German language allows (e.g., no separate word for many smaller islands, e.g., Isle of Man, Curaçao). Assisted by the preeminent German language dictionary [duden.de](http://www.duden.de).

Many existing dictionaries

You find a list at

<https://meteor.opted.eu/>

Meteor
Media Texts Open Registry v1.2.2

[Login](#) [Register](#) [Resources](#) [About](#)

Search everything...

Total results: 14

Name	Type	Country
ConText Diesner, J et al. (2020)	Tool	
DDR Garten, Justin et al. (2017)	Tool	
DICTION Roderick P. Hart (1996)	Tool	
LIWC Pennebaker, J. W. et al. (1999)	Tool	
NLTK NLTK Team (2001)	Tool	
Netlytic Gruz, A. (2016)	Tool	
T-LAB T-LAB di Lancia Franco	Tool	
WordStat Provalis Research	Tool	
corpustools Welbers K et al. (2018)	Tool	
iLCM Andreas Niekler et al. (2018)	Tool	
popdictR Gründl, Johann (2020)	Tool	
quanteda Benoit, Kenneth et al. (2018)	Tool	
tidytext De Queiroz, Gabriela et al. (2016)	Tool	
tm Feinerer, Ingo et al. (2008)	Tool	

Free text search

Entity Type

Tool x

Countries ☐ and ☐ or

Channel

Languages ☐ and ☐ or

Used For ☐ and ☐ or

Dictionary Analysis x

Concept Variables ☐ and ☐ or

Programming Languages ☐ and ☐ or

Dictionaries = Search strings?

- When dictionaries are used to search for data in a repository they are called search strings

Table 3 Boolean search strings used for retrieval of migration-related news articles

Country	Language	Search string
Spain	Spanish	asilo* OR inmigra* OR refugiad* OR migrante* OR migratori* OR "sin papeles" OR "campo de desplazados" OR patera* OR emigra* OR "libre circulación" OR "fuga de cerebros"
UK	English	asyl* OR immigrant* OR immigrat* OR migrant* OR migrat* OR refugee* OR foreigner* OR "undocumented worker*" OR "guest worker*" OR "foreign worker*" OR emigrat* OR "freedom of movement" OR "free movement"
Germany	German	asyl* OR immigrant* OR immigriert* OR immigrat* OR migrant* OR migrat* OR flüchtling* OR ausländer* OR zuwander* OR zugewander* OR einwander* OR eingewander* OR gastarbeiter* OR "ausländische arbeitnehmer*" OR emigr* OR auswander* OR ausgewander* OR personenfreizügigkeit* OR arbeitnehmerfreizügigkeit* OR "freier personenverkehr*"

Validate, validate, validate

Table 1 Four principles of quantitative text analysis

-
- (1) All quantitative models of language are wrong—but some are useful.
 - (2) Quantitative methods for text amplify resources and augment humans.
 - (3) There is no globally best method for automated text analysis.
 - (4) Validate, Validate, Validate.
-

Dictionary validation

- Face validity: Do terms intuitively make sense?
 - External validity: Compare dictionary scores against manual codes or other external measures.
-

External validity

- **Construct validity:** How close is an automated measurement to a more trusted measurement
- The more trusted measurement is **typically human understanding of text**
 - Why? As social scientists, we typically care about how readers understand a text, or what communicators intended to say

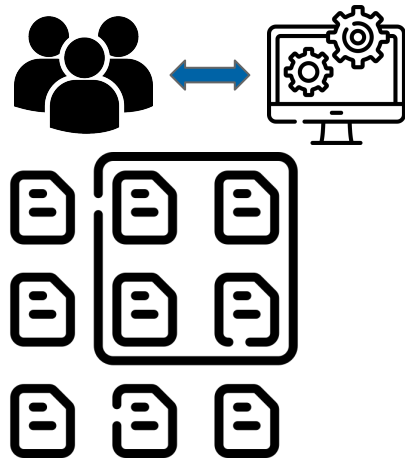


Validation with manually created baseline

Steps

1. Select a subset of the documents
2. Label the subset manually
3. Classify the subset automatically (with dictionary)
4. Compare manual decisions with automated decisions (via metrics calculation: recall, precision, F1)

Low metrics? → improve the dictionary and calculate metrics again



Creation of a manual baseline

- Codebook: Rules, Examples
- Coder Selection: Expert coders vs. Crowdcoders; Coder Training
- Quality Assessment: Inter-coder reliability, majority vote, *valid disagreement* (Baden et al., 2023)
- Document Selection: Representative for target discourse (e.g., random selection or artificial week)

Baden, C., Boxman-Shabtai, L., Tenenboim-Weinblatt, K., Overbeck, M., & Aharoni, T. (2023). Meaning multiplicity and valid disagreement in textual measurement: A plea for a revised notion of reliability. *SCM Studies in Communication and Media*, 12(4), 305-326.

Lind, F., Gruber, M., & Boomgaarden, H. G. (2017). Content analysis by the crowd: Assessing the usability of crowdsourcing for coding latent constructs. *Communication methods and measures*, 11(3), 191-209.

Tools for manual coding

- Google sheets
 - Excel
 - Soscisurvey <https://www.soscisurvey.de/>
 - Label Studio: <https://labelstud.io/academic/>
-

Recall, precision?

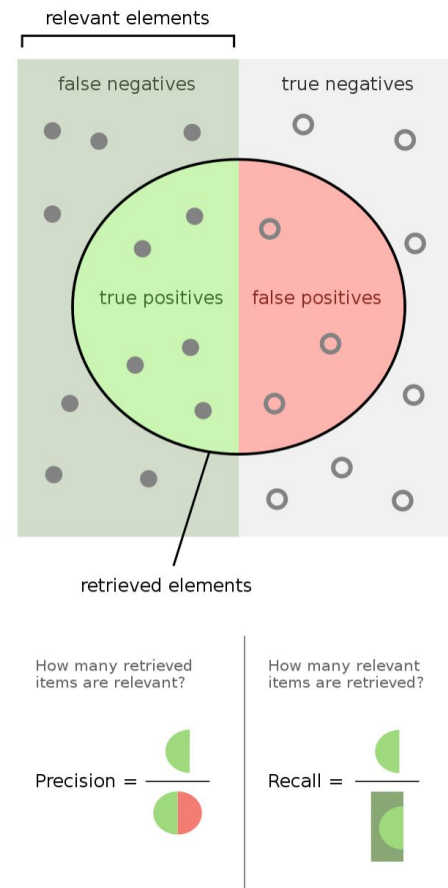
Metrics frequently used to express the **validity** of a dictionary & other automated classification methods

Range: 0 (lowest) - 1 (highest)

Example category: Inflation (yes/no)

- **Recall:** Of all the actual texts about “Inflation” cases, how many did the model find?
- **Precision:** Of all the times the model predicted “Inflation”, how often was it correct?

Scharkow, M. (2011). Zur Verknüpfung manueller und automatischer Inhaltsanalyse durch maschinelles Lernen. *M&K Medien & Kommunikationswissenschaft*, 59(4), 545-562.



F1

The F1-Score is the harmonic mean of Precision and Recall. It provides a single, balanced score.

$$F1 = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$$

What to report? F1 alone is not sufficient, can hide problems with recall or precision

Summary Dictionaries

Pros

- Often needed to select data (search strings)
- High reliability and control
- High transparency and reproducibility

Cons

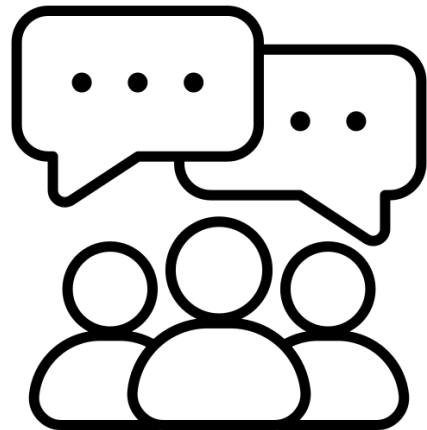
- Difficulty increases with the latency of the construct
 - May miss language nuances
-

What do we see here?

"#\w+"

"\\b\\d{4}\\b"

"(?<=Speaker:)\\b[A-Z][a-z]+\\s[A-Z][a-z]+"



What do we see here?

Regular expressions:

"#\w+" Extracts hashtags, often used for social media analysis

"\\b\\d{4}\\b" Extracts 4-digit numbers, often used to find years.

"(?<=Speaker:)\\b[A-Z][a-z]+\\s[A-Z][a-z]+" Extracts speaker names following "Speaker: " in text.

Regular Expressions (regex)

- Formal language to specify search strings



Regular Expressions (regex)

- Formal language to specify search strings
 - Insanely difficult
-

Regular Expressions (regex)

- Formal language to specify search strings
- INSANELY difficult



Regular Expressions (regex)

- Formal language to specify search strings
- ***INSANELY*** difficult



Regular Expressions (regex)

- Formal language to specify search strings
 - **INSANELY** difficult
 - Nobody can remember anything
-

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 - Different **flavours**
-

Regular Expressions (regex)

- Formal language to specify search strings
 - **INSANELY** difficult
 - Nobody can remember anything
 - Different **flavours**
-
- “Some people, when confronted with a problem, think “I know, I’ll use regular expressions.” Now they have two problems.” *Jamie Zawinski*
-

How difficult to regex an email

How difficult to regex an email

Rather

How difficult to regex an email

```
(?:[a-z0-9!#$%&'*/+=?^_`{|}~-]+(?:\. [a-z0-9!#$%&'*/+=?^_`{|}~-]+)*|"(?:[\x01-  
\x08\x0b\x0c\x0e-\x1f\x21\x23-\x5b\x5d-\x7f]|\\[\x01-\x09\x0b\x0c\x0e-\x7f])*")@(?:  
(?:[a-z0-9](?:[a-z0-9-]*[a-z0-9])?\.)+[a-z0-9](?:[a-z0-9-]*[a-z0-9])?|\[(?:(?:(2(5[0-  
5]|[0-4][0-9]))|1[0-9][0-9]|[1-9]?[0-9]))\\.){3}(?: (2(5[0-5]|[0-4][0-9]))|1[0-9][0-9]|[1-  
9]?[0-9])|([a-z0-9-]*[a-z0-9]:(?:[\x01-\x08\x0b\x0c\x0e-\x1f\x21-\x5a\x53-\x7f]|\\  
[\x01-\x09\x0b\x0c\x0e-\x7f]))+)\])
```

[illegible]

[illegible]

- Disjunctions

Grimmer / Jurafsky
Cheat-sheet

RE	Match	Example Patterns Matched
[mM]oney	Money or money	" <u>M</u> oney"
[abc]	'a', 'b', <i>or</i> 'c'	"Investing in Ir <u>a</u> n" "is d <u>a</u> ngerous <u>b</u> usiness"
[1234567890]	any digit	"sitting on \$ <u>7</u> . <u>5</u> billion dollars" " <u>2005</u> and <u>2006</u> , more than " "\$ <u>150</u> million dollars"
[\.]	A period	" 'Run!', he screamed <u>.</u> "

- Ranges

RE	Match	Example Patterns Matched
[A-Z]	an upper case letter	" <u>R</u> ep. <u>A</u> nthony <u>W</u> einer (<u>D</u> - <u>B</u> rooklyn & <u>Q</u> ueens)"
[a-z]	a lower case letter	"ACORN' <u>s</u> "
[0-9]	a single digit	"(<u>9</u> th CD) "

- Negations

RE	Match	Example Patterns Matched
[^A-Z]	not an upper case letter	"ACORN <u>s</u> "
[^Ss]	neither 'S' nor 's'	" <u>ACORN</u> 's"
[^\.]	not a period	" <u>'Run!'</u> , he screamed."

- Optional Characters: ?, *, +

RE	Match	Example Patterns Matched
colou?r	Words with u 0 or 1 times	" <u>color</u> " or " <u>colour</u> "
oo*h!	Words with o 0 or more times	" <u>oh!</u> " or " <u>ooh!</u> " or " <u>oooh!</u> "
o+h!	Words with o 1 or more times	" <u>oh!</u> " or " <u>ooh!</u> " or " <u>oooooh!</u> " or

- Start of the line anchor `^`, end of the line anchor `$`

RE	Match	Example Patterns Matched
<code>^[A-Z]</code>	Upper case start of line	<code>"<u>P</u>alo Alto"</code> <code>"the town of Palo Alto"</code>
<code>^[^A-Z]</code>	Not upper case start of line	<code>"<u>t</u>he town of Palo Alto"</code> <code>"Palo Alto"</code>
<code>^.</code>	Start of line	<code>"<u>P</u>alo Alto"</code> <code>"<u>t</u>he town of Palo Alto"</code>
<code>.\$</code>	Identify character that ends a line	<code>"Wait<u>!</u>"</code> <code>"This is the end<u>.</u>"</code>

- "Or" | statements, Useful short hand

RE	Match	Example Patterns Matched
<code>yours mine</code>	Matches "yours" or "mine"	<code>"it's either <u>yours</u> or <u>mine</u>"</code>
<code>\ d</code>	Any digit	<code>"<u>1</u>-Mississippi"</code>
<code>\ D</code>	Any non-digit	<code>"<u>1</u>-Mississippi"</code>
<code>\ s</code>	Any whitespace character	<code>"<u>1, 2</u>"</code>
<code>\ S</code>	Any non-whitespace character	<code>"<u>1, 2</u>"</code>
<code>\ w</code>	Any alpha-numeric	<code>"<u>1-Mississippi</u> "</code>
<code>\ W</code>	Any non-alpha numeric	<code>"<u>1-Mississippi</u>"</code>

Helpers

GPT, Claude, etc. are great with regex

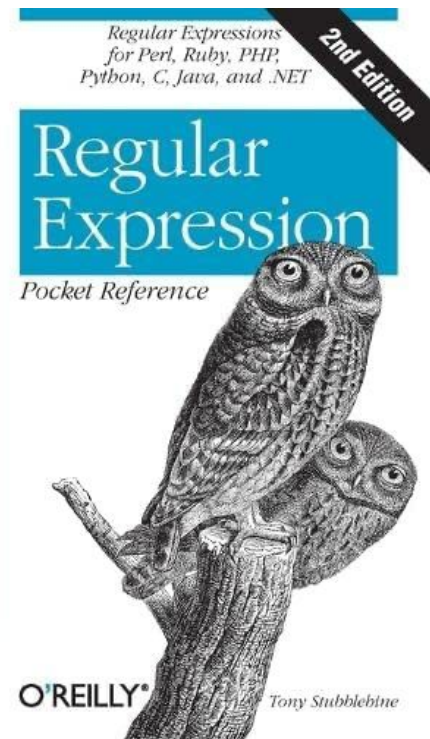
<https://regex101.com>

Recommended R tutorial (Wickham & Grolemund, 2017):

<https://r4ds.had.co.nz/strings.html>

To test regular expressions quickly:

https://spannbaueradam.shinyapps.io/r_regex_tester/



Coding session

Jointly:

1. Use an existing dictionary:
sentiment_dictionary.Rdm
LSDprep_dec2017.R
2. Create a dictionary + validate it:
dictionary_creation_and_validation.Rmd

Get scripts:

<https://github.com/fabien nelind/text-as-data-in-R>

Supervised classification

Supervised machine learning

Objectives:

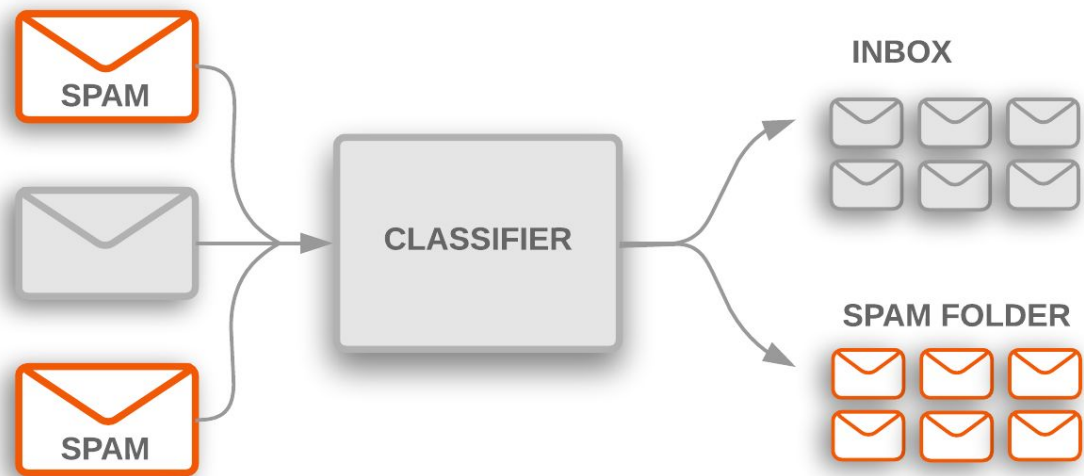
- Classification (for categorical variables)
 - E.g.: classify documents into pre existing categories
- Regression (for continuous variables)

Supervised machine learning

Objectives:

- **Classification (for categorical variables)**
 - **E.g.: classify documents into pre existing categories**
 - Regression (for continuous variables)
-

Email classification



Steps

1. Create a labeled data set
2. Training phase
3. Check performance
4. Use the model

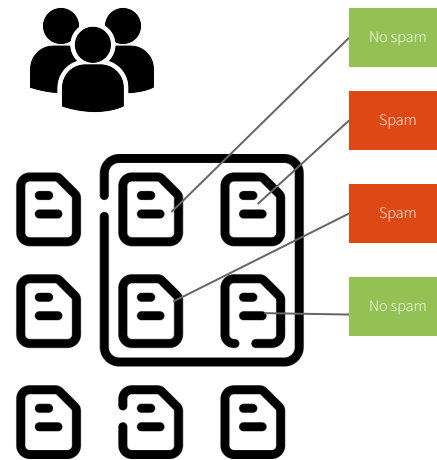
1. Create a labeled data set

How:

- Typically: Human coders label parts of the corpus

Considerations:

- Sampling should be representative for the corpus (e.g., Random, Stratified sample e.g., across time and source)
- Quality of human coding matters (intercoder reliability)
- Number of documents



1. Create a labeled data set

Split labeled data in training data, test data, validation set

Training data

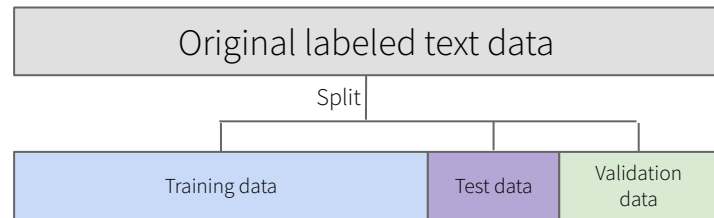
- The subset that is used to learn the model parameters

Test data

- Another subset used to evaluate the model's predictive quality
- Not used for learning!

Validation data

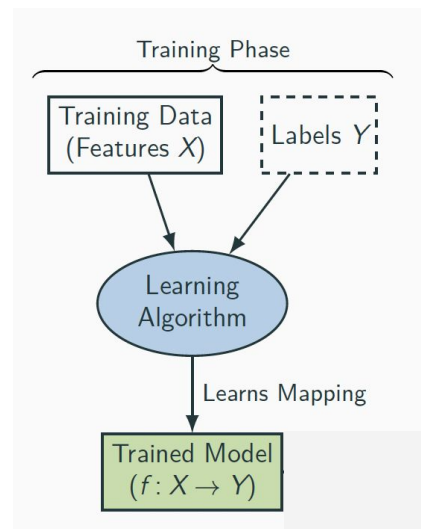
- Only used to evaluate in the end, no further optimization allowed



2. Training phase

Classifier learns the mapping between (text) features and the labels in the training set

- We define a model $f(Y)=g(X)$
- And apply a learning algorithm to establish which features in X (features extracted from the training documents) matter to recover Y (i.e, the labels of the training documents)
- We fit the model



2. Training phase

Considerations:

- Feature representation: Bag of words representation or embeddings
- Feature selection: Remove irrelevant features
- Classifier selection: e.g., Naive Bayes, SVM, KNN, or ensemble methods

2. Training phase

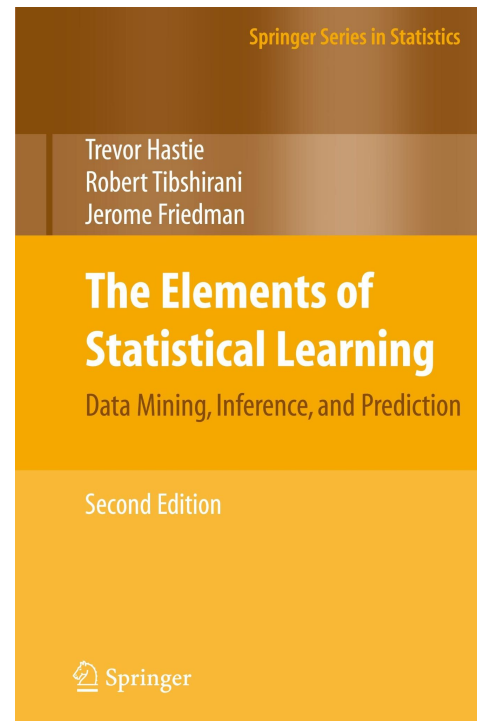
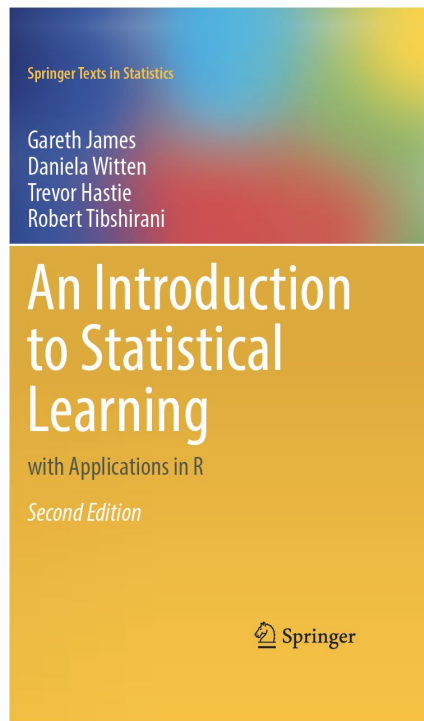
Naïve Bayes Classifier

- Probabilistic classifier
- Simple
- Fast
- Good Accuracy

Support Vector Machine

- Comes from computer science
 - Very good
 - Rather difficult math
 - Considered one of the best of-the-shelf classification algorithms
-

More details:



Performing supervised machine learning in R and Python

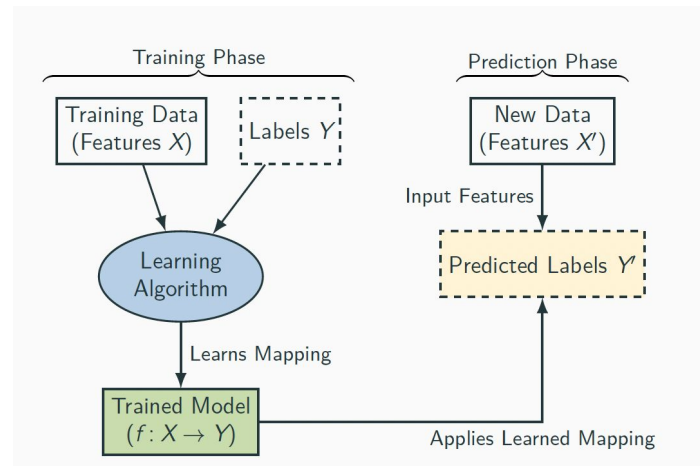
The classifiers are implemented in many stats/ML packages

R: quanteda, caret, e1071, klaR, C5.0, OneR

Python: scikit-learn

3. Check performance

The fitted model (the trained classifier) is applied to a held-out test set (was not used for training the model).



3. Check performance

- Compare predicted labels with manual labels for the **test set**
 - Inspect Confusion matrix
 - Calculate recall, precision, F1
- Bad results for test set: Improve training phase
- Good results for test set: Repeat for **validation set**

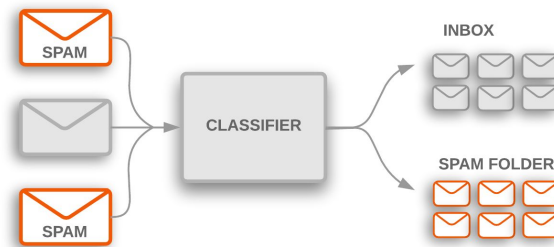
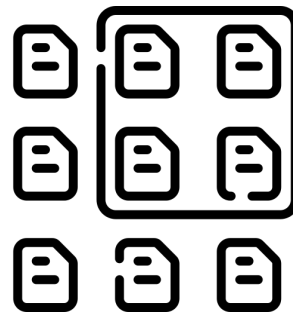
Confusion matrix

Classification (algorithm)	Actual label	
	Negative	Positive
Negative	True negative	False negative
Positive	False negative	True positive

4. Using the model

Remember that we labeled only parts of our original corpus.

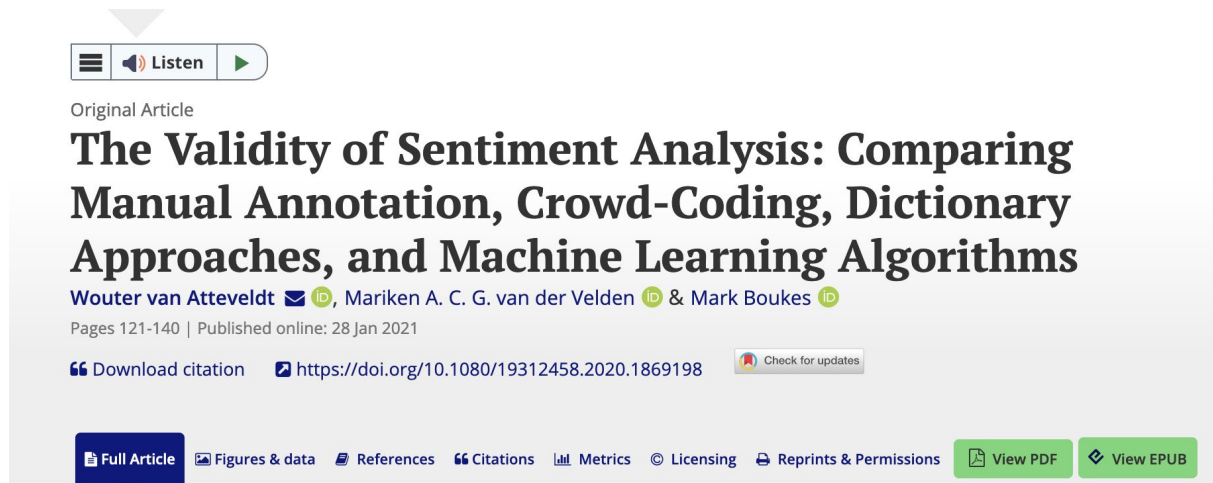
When you are satisfied with the performance of the classifier, you can use it to classify all documents in the corpus



Dictionary vs. supervised machine learning

Category: Sentiment

Result: machine learning significantly outperformed dictionary coding



The screenshot shows the top portion of a research article page. At the top, there is a 'Listen' button with a speaker icon. Below it, the text 'Original Article' is displayed. The main title of the article is 'The Validity of Sentiment Analysis: Comparing Manual Annotation, Crowd-Coding, Dictionary Approaches, and Machine Learning Algorithms'. The authors listed are Wouter van Atteveldt, Mariken A. C. G. van der Velden, and Mark Boukes, each with a small ORCID icon. Below the authors, it says 'Pages 121-140 | Published online: 28 Jan 2021'. There are links for 'Download citation' and the DOI 'https://doi.org/10.1080/19312458.2020.1869198'. A 'Check for updates' button is also present. At the bottom of this section, there is a row of buttons: 'Full Article' (highlighted in dark blue), 'Figures & data', 'References', 'Citations', 'Metrics', 'Licensing', 'Reprints & Permissions', 'View PDF', and 'View EPUB'.

  Listen 

Original Article

The Validity of Sentiment Analysis: Comparing Manual Annotation, Crowd-Coding, Dictionary Approaches, and Machine Learning Algorithms

Wouter van Atteveldt  , Mariken A. C. G. van der Velden  & Mark Boukes 

Pages 121-140 | Published online: 28 Jan 2021

 Download citation  <https://doi.org/10.1080/19312458.2020.1869198>  Check for updates

 Full Article  Figures & data  References  Citations  Metrics  Licensing  Reprints & Permissions  View PDF  View EPUB

ABSTRACT

Van Atteveldt et al. (2021)

Dictionary vs. supervised machine learning

- Supervised machine learning requires (potentially larger amounts) of labeled data
- If the training sample is large enough supervised learning will outperform dictionaries

Coding session

- Sentiment
in movie
reviews



I liked it, but thought the third act nearly cratered the whole thing.

January 3, 2024 | [Full Review...](#)

Coding session

- Sentiment in movie reviews



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An intense, inventively filmed, and well-acted biopicture about the kinds of events that are hard on the heart.

[Full Review](#) | Original Score: 5/5 | Nov 20, 2023



In a movie about impending global catastrophe, he gives a close-up of a face, and a twitch of a lip the power of an atom bomb.

[Full Review](#) | Original Score: A | Nov 17, 2023



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[Full Review](#) | Original Score: A | Nov 17, 2023

Coding session

Jointly:

Apply and validate classifier

supervised_nb_svm.RmD

Get script:

<https://github.com/fabien nelind/text-as-data-in-R>

What is next on the program

MIDDAY

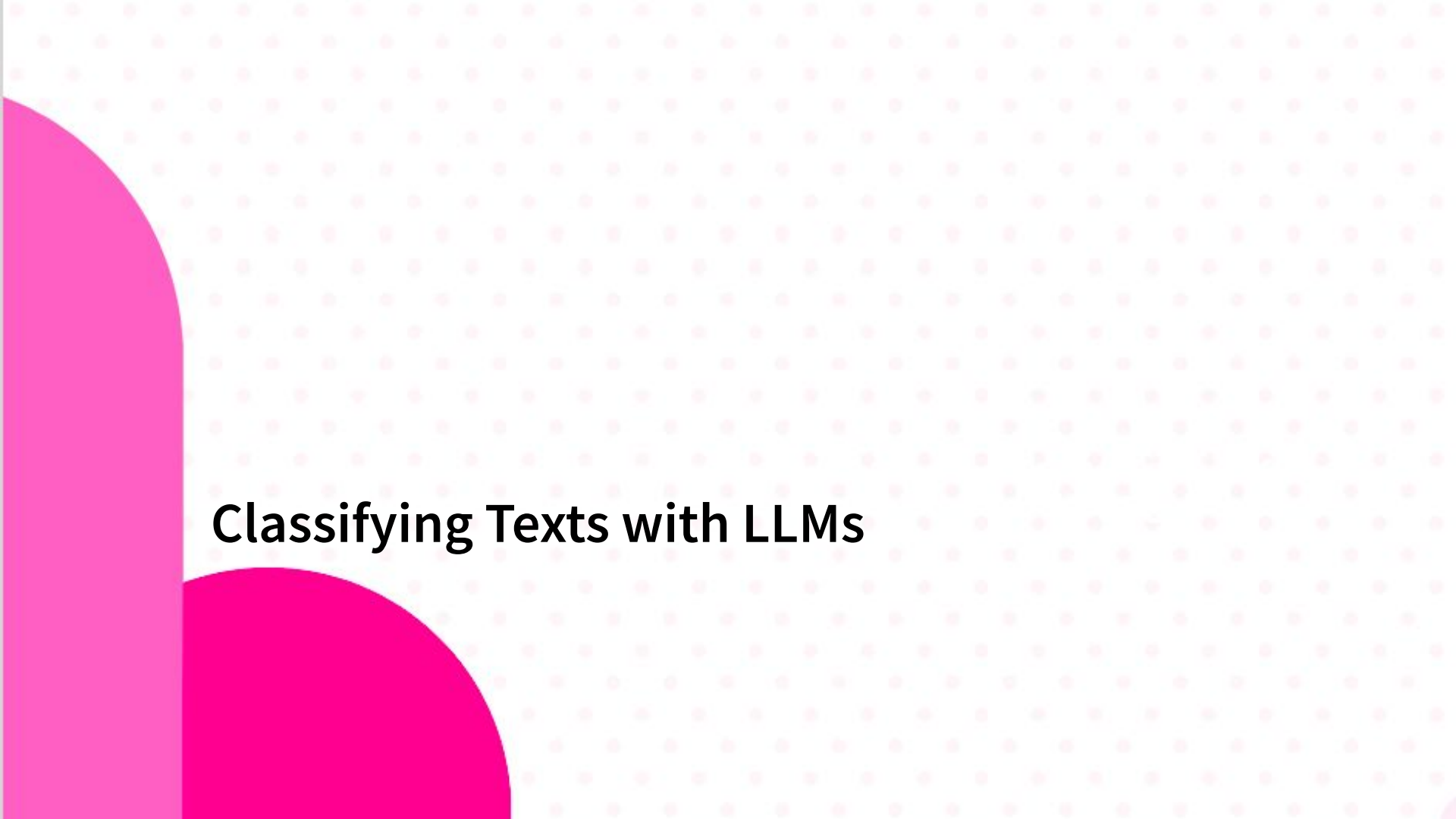
12:30

13:30

LUNCH BREAK

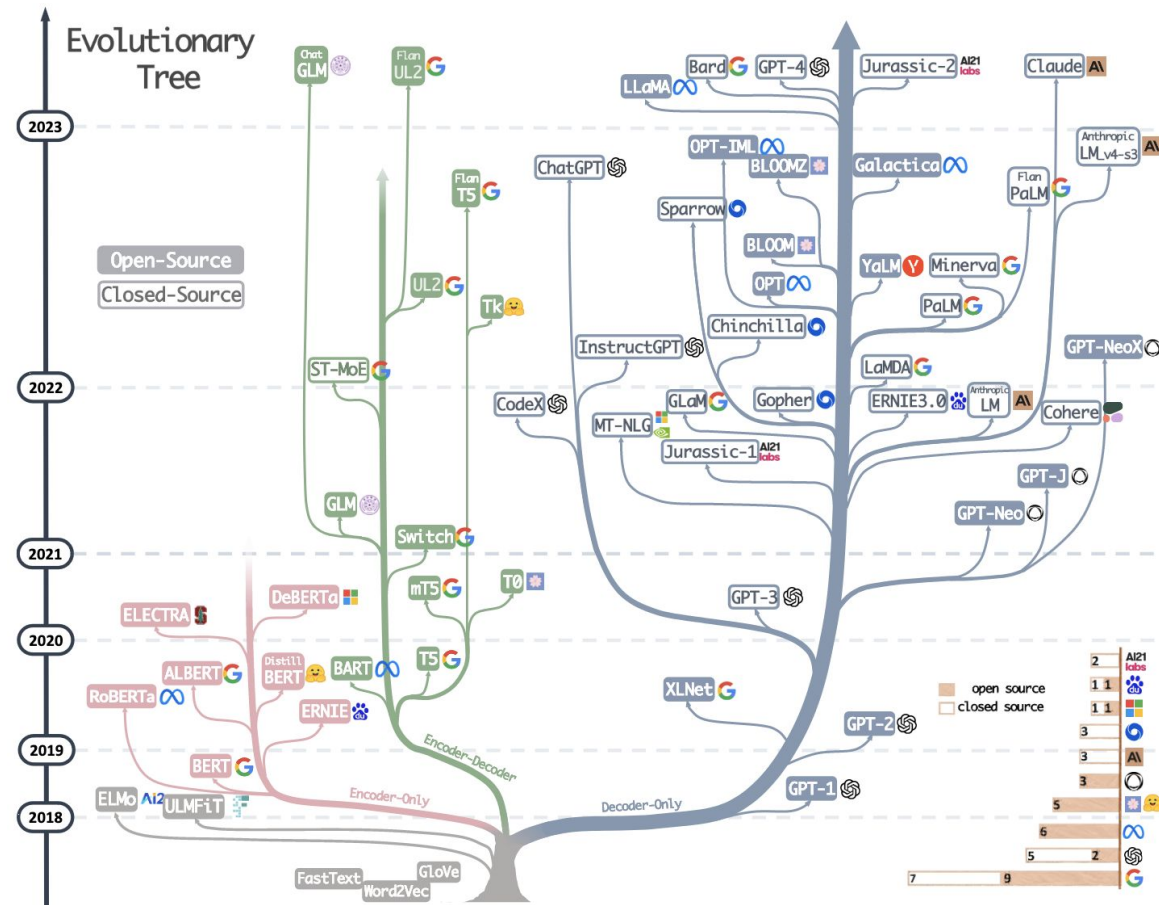
MZES

A5, 6, BUILDING A,
68159 MANNHEIM

The background features a light pink grid of small dots. On the left side, there are two large, overlapping magenta shapes: a vertical bar with a rounded right edge and a semi-circle below it.

Classifying Texts with LLMs

Model evolution



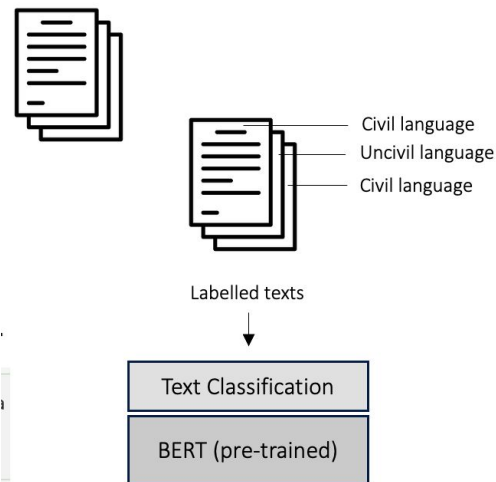
Yang, J., H. Jin, R. Tang, X. Han, Q. Feng, H. Jiang, ... and X. Hu. 2023. "Harnessing the Power of LLMs in Practice: A Survey on ChatGPT and Beyond." *arXiv preprint arXiv:2304.13712*.

Advancements in Text Classification Using Transformer Architecture

Researchers increasingly harness the power of transformer architecture for text classification tasks due to its high performance.

Main Strategies (Bosley et al., 2023):

- **fine-tuning smaller LLMs (e.g., BERT)**



Advancements in Text Classification Using Transformer Architecture

Researchers increasingly harness the power of transformer architecture for text classification tasks due to its high performance.

Main Strategies (Bosley et al., 2023):

- fine-tuning smaller LLMs (e.g., BERT)
- **querying larger LLMs (e.g., GPT-3)**



```
prompt = "Does the following text include uncivil language? Answer only with a  
number: 0 or 1. 0 means No 1 means Yes. Output only the number."
```

Some differences

Fine-Tuning (e.g., BERT)

Goal: Teach the model a new skill

Method: Provide thousands of labeled examples to update the internal weights of the model via backpropagation

Model size: BERT-base 110 million parameters

In-Context Learning (e.g., GPT)

Goal: Instruct the model to perform a task

Method: Provide instructions (and a few examples) within the prompt - the model's weights are not updated.

Model size: GPT-4 has more than 1 trillion parameters

Different classification methods in comparison

	Classic supervised machine learning	Fine-tuning smaller LLMs (e.g., BERT)	Querying larger LLMs (e.g., GPT-3)
1.	Create a labeled data set (train, test, validate)	Create a labeled data set (train, test, validate)	Create a labeled data set (test, validate)
2.	Classify documents with supervised learning algorithm (e.g., SVM, Naive bayes)	- Take a pre-trained transformer model - Add classification layer on top of output embeddings - Fine-tuning: take labeled text dataset to train Transformer+classifier through supervised learning	-Query the model by providing the instruction (codebook + text) as prompt -Experiment with prompts and set-up
3.	Check performance	Check performance	Check performance
4.	Use the model for the full corpus	Use the model for the full corpus	Use the model for the full corpus

How to implement: Fine-tuning smaller LLMs (e.g., BERT)

Not covered in this
workshop, but see:



Applied examples:

- identify populist campaign messages (Bonkowski et al., 2022) <https://doi.org/10.1177/004912412211223>
- categorize the content of social (and other) media (Kroon et al., 2023)
<https://doi.org/10.1080/19312458.2023.2261372>

Python:

- Learn Python for Everybody: <https://www.youtube.com/watch?v=8DvywoWv6fI>
- Data Analysis with Python: <https://www.youtube.com/watch?v=GPVsHOIRBBI>
- VanderPlas, J. (2016). Python Data Science Handbook: Essential Tools
<https://github.com/jakevdp/PythonDataScienceHandbook>

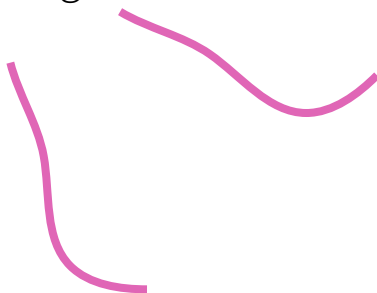
Tutorial: <https://huggingface.co/docs/transformers/training>

https://lena-voita.github.io/nlp_course/seq2seq_and_attention.html

How to implement: querying larger LLMs (e.g., GPT-3):

Let's get started!

We run through 2 examples together

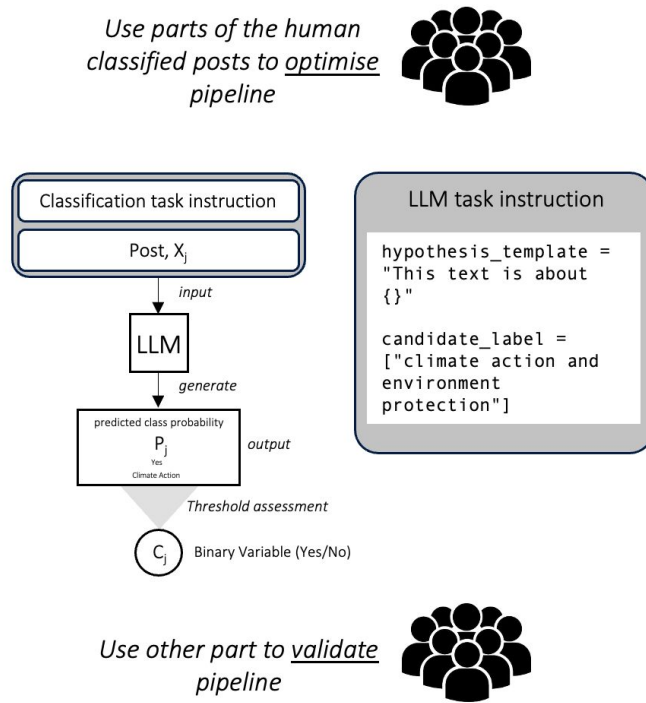


Run a commercial
model via API

Run an open-source
model on your laptop

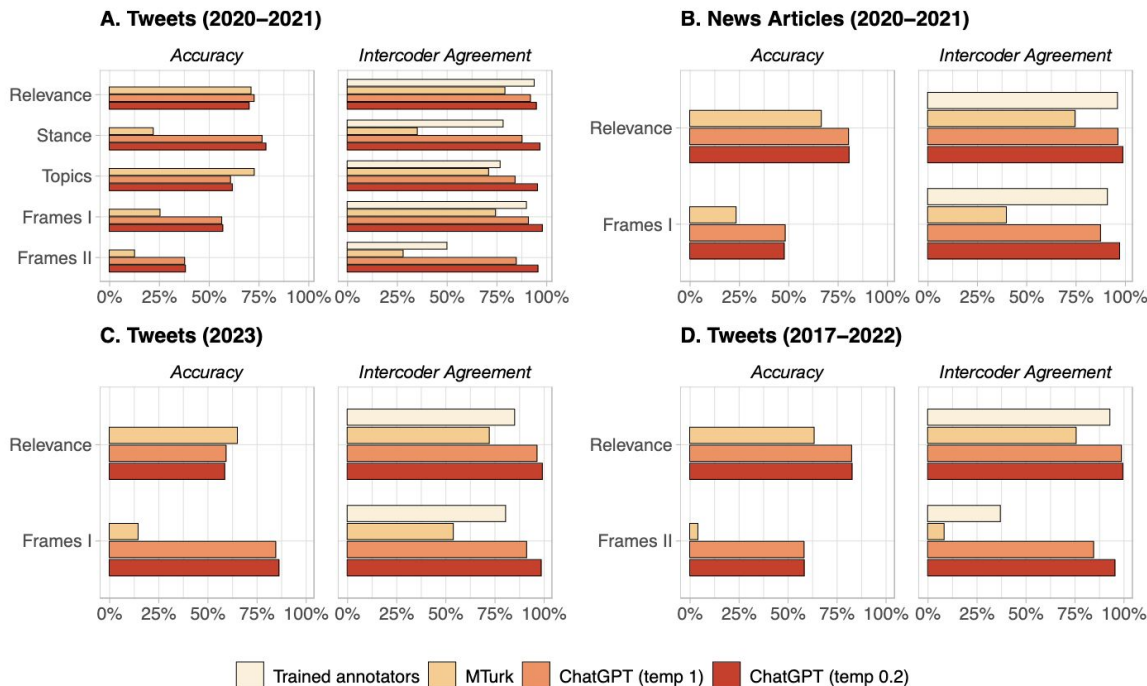
Using LLMs for text annotation

Task: Perform topic analysis on a social media post



GPT for text annotation

- Testing GPT for several annotation tasks, including relevance, stance, topics, and frames detection



LLMs beyond classification

Research Assistant:

- a. **Summarisation:** Condense scientific papers, long policy reports, or legal texts
- b. **Information Extraction:** Pull out predefined fields or facts from unstructured text.
 - i. Prompt: “Identify all direct quotes in the article and attribute them. For each quote, extract: speaker_name” (see Wozniak et al., 2026)
- c. **Augmentation:** Generate synthetic examples for specific categories to help train or fine-tune smaller, domain-specialized models.

LLMs beyond classification

Studying LLMs:

- a. **Probing for bias:** Systematically query models to understand the social, political, cultural biases they have.
 - i. Prompt: “List 10 public holidays” (Wang et al., 2024)
- b. **Simulation:** Can LLMs be prompted to act as survey respondents? (Argyle et al., 2023; Bisbee et al., 2024)
- c. **Human-AI Interaction:** How do people interact with, trust, and get influenced by LLMs?

Ethics: No Neutral Models

- LLMs learn from web-scale text, and inherit the social biases embedded in that data

Implications (Examples):

- Sentiment classification across demographic categories may exhibit measurable bias
- If you generate “mock” interview data, you may get stereotyped language and missing nuance for underrepresented groups.

➔ Treat LLM outputs as hypotheses requiring validation, not ground truth

Ethics: Privacy, Equity, and Footprint

- **Privacy:** LLMs can leak personal data; sending sensitive research data to APIs poses privacy risks (e.g., interview transcripts, private social media chats)
- **Access divide and control:** Powerful LLMs are often proprietary and controlled by a few large tech companies; who has access? Who decides?
- **Environment:** Training and also prompting a single large LLM consume substantial energy and water

Data
Management
Plan

Check Footprint:
<https://calculator.ecologits.ai/>

Parameters to consider

- **Model:** Matters for performance (different model sizes, different input formats, different output formats, different languages etc.)
 - **Temperature:** A measure of how often the model outputs a less likely token. The higher the temperature, the more random (and usually creative) the output. For most factual use cases such as data extraction, and truthful Q&A, the temperature of 0 is best.
-

Considerations

Costs:

Estimate Costs:
<https://gptforwork.com/tools/openai-chatgpt-api-pricing-calculator>

Example, no promotion

Flagship models								
Our latest models								
Prices per 1M tokens.								
Model	Short context				Long context			
	Input	Cached input	Cache writes	Output	Input	Cached input	Cache writes	Output
gpt-5.6-sol	\$5.00	\$0.50	\$6.25	\$30.00	\$10.00	\$1.00	\$12.50	\$45.00
gpt-5.6-terra	\$2.50	\$0.25	\$3.125	\$15.00	\$5.00	\$0.50	\$6.25	\$22.50
gpt-5.6-luna	\$1.00	\$0.10	\$1.25	\$6.00	\$2.00	\$0.20	\$2.50	\$9.00

Reproducibility: proprietary models (e.g., GPT) are not open-source and hinder reproducibility (Spriling, 2023)

Replicability: even minor wording alterations in prompts or repeating the identical input can lead to varying outputs (Reiss, 2023)

To run commercial models locally

Install:

1. R (and optionally RStudio)
2. R packages: `install.packages(c("httr", "jsonlite"))`
3. Create a developer account and API key
 - Provider (for example): OpenAI
<https://platform.openai.com/>
 - Create an API key (User menu → View API Keys
→ Create new secret key)
 - Add billing and set a low hard/soft usage limit
for safety (Usage → Limits)

Consider:

1. **Store your API key securely**

Do NOT hardcode in scripts.

2. **Reproducibility:** Record model names and versions used

3. **Privacy:** Don't send sensitive or personal data to external APIs unless approved.



To run Ollama models locally

Install:

1. R (and optionally RStudio)
2. R packages: `install.packages(c("rollama", "tidyverse"))`
3. Ollama (the local LLM runtime that rollama talks to)
 - Download and install from <https://ollama.com/download>

Consider:

1. **Disk space:** the 3B q8_0 model for example is roughly 2–3 GB download. Ensure at least ~5 GB free.
2. **RAM:** 3B q8_0 runs comfortably on 8–16 GB RAM systems; more RAM helps.
3. **CPU/GPU:** Ollama runs on CPU by default and can optionally use GPU.

Coding session

Jointly:

- GPT_prompting.Rmd
- ollama_classification.R

Get script:

<https://github.com/fabien nelind/text-as-data-in-R>

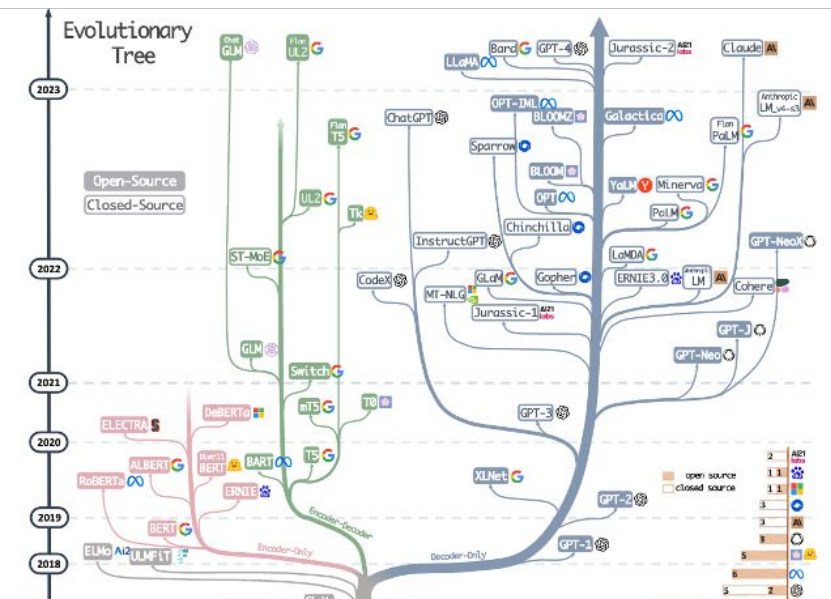
The background features a light pink grid of small dots. On the left side, there are two large, solid pink shapes: a vertical bar with a rounded right edge and a semi-circle below it.

Wrap-Up

Central decision criteria

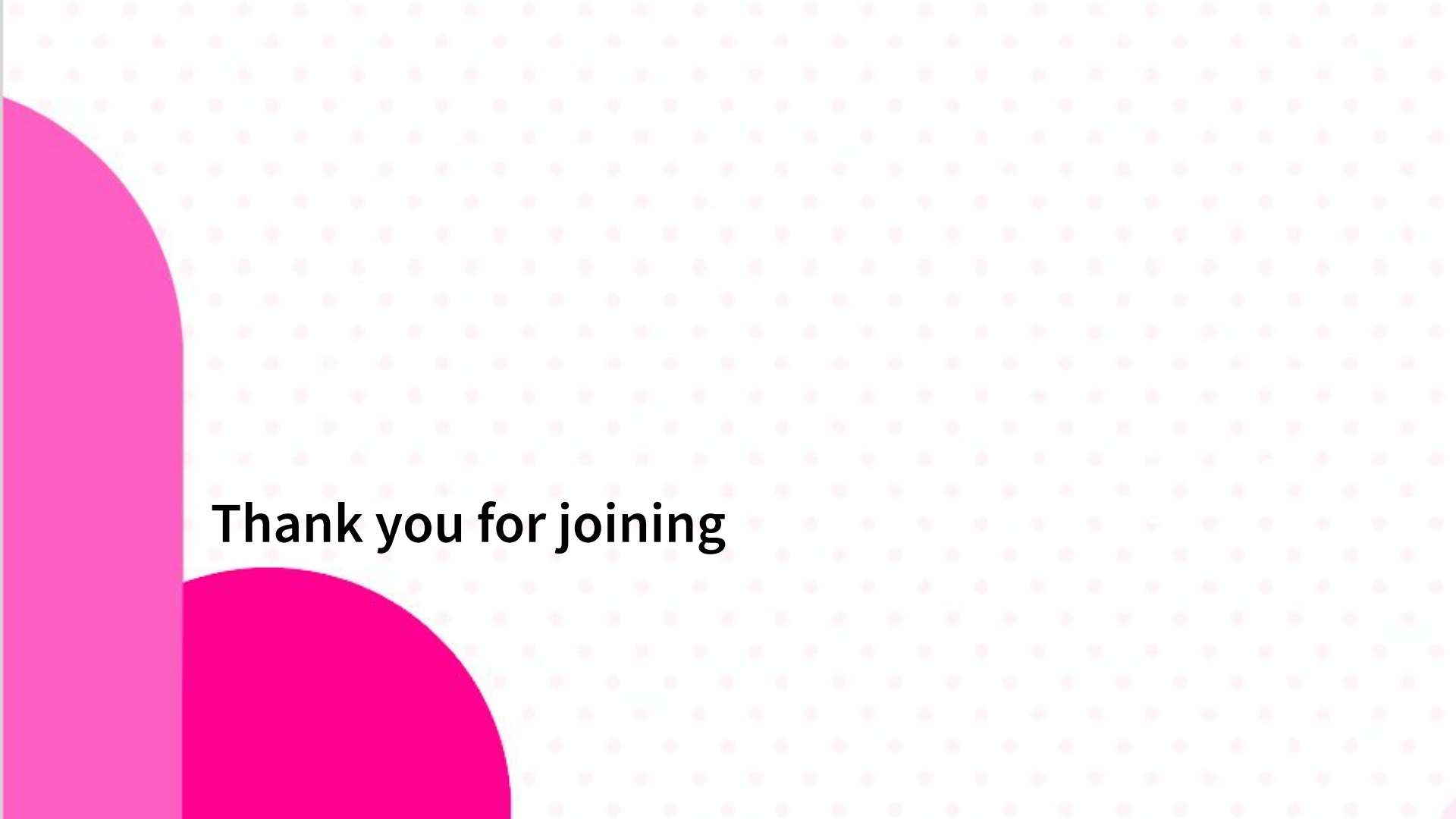
- Approach to measurement (discovery or finding pre-defined concept)?
 - Availability of methods (dictionaries, labeled data, pretrained models)
 - Time (and Importance for a specific RQ)
 - Budget (Modeling, Validation)
 - Skills (R, Python, Patience)
 - Ethical and environmental considerations
 - ...
-

Conclusion



Yang et al., 2023, p. 3

- The ways that we conduct text analysis is changing drastically, but questions that we ask about validation do not
- Social scientists with computational training can help to ask the interesting questions, curate datasets, can ensure valid implementation of methods, and interpret the results

The background is white with a grid of small, light pink dots. On the left side, there are two overlapping magenta shapes: a large semi-circle at the top and a smaller semi-circle below it.

Thank you for joining